CATERPILLAR®

DEALER BEST PRACTICE SERIES

Cost Reduction Through Haul Road Management Process

Application Maintenance Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

Poor haul road conditions are a commonly cited cause of accelerated machine wear, operator fatigue, and mine inefficiency. These factors can increase risk in MARC contracts and severely effect profitability if not kept in check.

MARC stakeholders frequently begin haul road defect management initiatives in response to adverse operating conditions and poor operator skill. Aiming to correlate the severity of machine application to component life, these studies usually investigate data from strut cylinders, transmissions, differentials, and final drives. Factors like tire life and frame life are also analyzed.

A proper road defect management process uses several tools, including:

- **Fleet Productivity Optimization (FPO) software:** A software tool designed to identify and localize potential problems in roads by using on-board electronic information from hauling trucks.
- **Fleet, Productivity and Cost analysis (FPC) software:** A software tool designed to assess changes in road design with productivity improvements and fuel savings.
- **Road Defect Management (RDM) methodology:** A methodology exclusive to Caterpillar (Patent Pending) that describes a preventative approach to maintain haul roads, comprising of the following elements:
    - Condition Monitoring
    - Road Repair processes
    - Plan and scheduling
    - Assessment of benefits

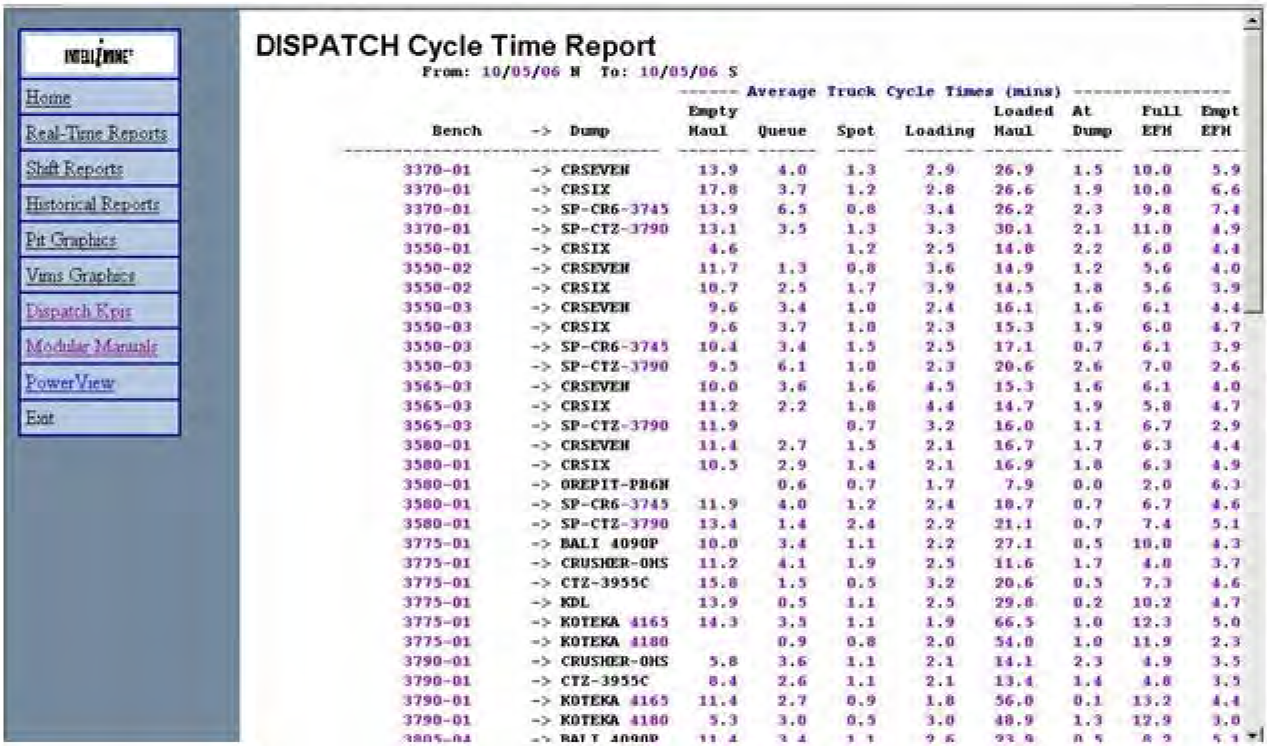
This process provides tangible benefits to both customers and dealers. Improving productivity and reducing overall operating costs can lower cost per ton, while minimizing haul road hazards will help safety. Additionally, dealers can benefit from reduced risk in MARCs and lower overall repair and maintenance costs.

2.0 Best Practice Description

The basis of a Road Defect Management process involves Condition Monitoring, operator training, road repair, and all associated follow-up. A dedicated Machine Application and Operations crew has the goal to capture, analyze, and assess allocation and operating activities to identify potential improvements in component life and productivity.

Current "as-is" processes mainly focus on the use of FPO as the main tool. A better process expands into other areas and utilizes other tools. In addition to FPO, the Dealer uses VIMS and its interface with their local Mine Controller software, which provides the following reports (see attached files, Courtesy of Freeport mine site):

- **Dispatch Cycle Time Report:** showing detailed cycle time information for specific roads.

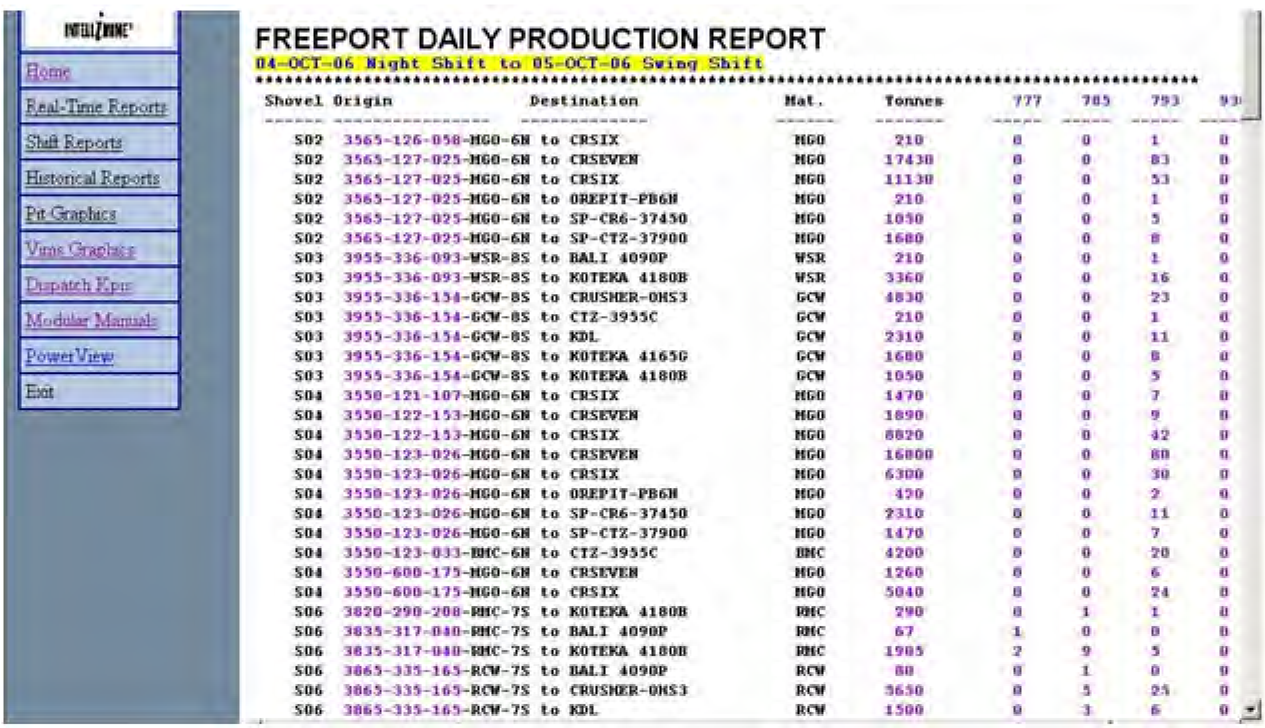


DISPATCH Cycle Time Report

From: 10/05/06 N To: 10/05/06 S

		Average Truck Cycle Times (mins)							
Bench	-> Dump	Empty Haul	Queue	Spot	Loading	Loaded Haul	At Dump	Full EFH	Empt EFH
3370-01	-> CRSEVEN	13.9	4.0	1.3	2.9	26.9	1.5	10.0	5.9
3370-01	-> CRSIX	17.8	3.7	1.2	2.8	26.6	1.9	10.0	6.6
3370-01	-> SP-CR6-3745	13.9	6.5	0.8	3.4	26.2	2.3	9.8	7.4
3370-01	-> SP-CTZ-3790	13.1	3.5	1.3	3.3	30.1	2.1	11.0	4.9
3550-01	-> CRSIX	4.6		1.2	2.5	14.0	2.2	6.0	4.4
3550-02	-> CRSEVEN	11.7	1.3	0.8	3.6	14.9	1.2	5.6	4.0
3550-02	-> CRSIX	10.7	2.5	1.7	3.9	14.5	1.8	5.6	3.9
3550-03	-> CRSEVEN	9.6	3.4	1.0	2.4	16.1	1.6	6.1	4.4
3550-03	-> CRSIX	9.6	3.7	1.0	2.3	15.3	1.9	6.0	4.7
3550-03	-> SP-CR6-3745	10.4	3.4	1.5	2.5	17.1	0.7	6.1	3.9
3550-03	-> SP-CTZ-3790	9.5	6.1	1.0	2.3	20.6	2.6	7.0	2.6
3565-03	-> CRSEVEN	10.0	3.6	1.6	4.5	15.3	1.6	6.1	4.0
3565-03	-> CRSIX	11.2	2.2	1.8	4.4	14.7	1.9	5.8	4.7
3565-03	-> SP-CTZ-3790	11.9		0.7	3.2	16.0	1.1	6.7	2.9
3580-01	-> CRSEVEN	11.4	2.7	1.5	2.1	16.7	1.7	6.3	4.4
3580-01	-> CRSIX	10.5	2.9	1.4	2.1	16.9	1.8	6.3	4.9
3580-01	-> OREPIT-PB6N		0.6	0.7	1.7	7.9	0.0	2.0	6.3
3580-01	-> SP-CR6-3745	11.9	4.0	1.2	2.4	18.7	0.7	6.7	4.6
3580-01	-> SP-CTZ-3790	13.4	1.4	2.4	2.2	21.1	0.7	7.4	5.1
3775-01	-> BALI 4090P	10.0	3.4	1.1	2.2	27.1	0.5	10.0	4.3
3775-01	-> CRUSHER-OHS	11.2	4.1	1.9	2.5	11.6	1.7	4.0	3.7
3775-01	-> CTZ-3955C	15.8	1.5	0.5	3.2	20.6	0.5	7.3	4.6
3775-01	-> KDL	13.9	0.5	1.1	2.5	29.8	0.2	10.2	4.7
3775-01	-> KOTEKA 4165	14.3	3.5	1.1	1.9	66.5	1.0	12.3	5.0
3775-01	-> KOTEKA 4180		0.9	0.8	2.0	54.0	1.0	11.9	2.3
3790-01	-> CRUSHER-OHS	5.8	3.6	1.1	2.1	14.1	2.3	4.9	3.5
3790-01	-> CTZ-3955C	8.4	2.6	1.1	2.1	13.4	1.4	4.8	3.5
3790-01	-> KOTEKA 4165	11.4	2.7	0.9	1.8	56.0	0.1	13.2	4.4
3790-01	-> KOTEKA 4180	5.3	3.0	0.5	3.0	48.9	1.3	12.9	3.0
3805-04	-> BALI 4090P	11.4	3.4	1.1	2.6	23.9	0.5	8.2	5.1

- **Daily Production Report:** showing Payload and Truck traffics for specific roads.



FREEPORT DAILY PRODUCTION REPORT

04-OCT-06 Night Shift to 05-OCT-06 Swing Shift

Shovel	Origin	Destination	Mat.	Tonnes	777	785	793	93
S02	3565-126-058-MGO-6N to	CRSIX	MGO	210	0	0	1	0
S02	3565-127-025-MGO-6N to	CRSEVEN	MGO	17430	0	0	83	0
S02	3565-127-025-MGO-6N to	CRSIX	MGO	11130	0	0	53	0
S02	3565-127-025-MGO-6N to	OREPIT-PB6N	MGO	210	0	0	1	0
S02	3565-127-025-MGO-6N to	SP-CR6-37450	MGO	1050	0	0	5	0
S02	3565-127-025-MGO-6N to	SP-CTZ-37900	MGO	1680	0	0	8	0
S03	3955-336-093-WSR-8S to	BALI 4090P	WSR	210	0	0	1	0
S03	3955-336-093-WSR-8S to	KOTEKA 4180B	WSR	3360	0	0	16	0
S03	3955-336-154-GCW-8S to	CRUSHER-OHS3	GCW	4830	0	0	23	0
S03	3955-336-154-GCW-8S to	CTZ-3955C	GCW	210	0	0	1	0
S03	3955-336-154-GCW-8S to	KDL	GCW	2310	0	0	11	0
S03	3955-336-154-GCW-8S to	KOTEKA 4165G	GCW	1680	0	0	8	0
S03	3955-336-154-GCW-8S to	KOTEKA 4180B	GCW	1050	0	0	5	0
S04	3550-121-107-MGO-6N to	CRSIX	MGO	1470	0	0	7	0
S04	3550-122-153-MGO-6N to	CRSEVEN	MGO	1890	0	0	9	0
S04	3550-122-153-MGO-6N to	CRSIX	MGO	8820	0	0	42	0
S04	3550-123-026-MGO-6N to	CRSEVEN	MGO	16800	0	0	80	0
S04	3550-123-026-MGO-6N to	CRSIX	MGO	6300	0	0	30	0
S04	3550-123-026-MGO-6N to	OREPIT-PB6N	MGO	420	0	0	2	0
S04	3550-123-026-MGO-6N to	SP-CR6-37450	MGO	2310	0	0	11	0
S04	3550-123-026-MGO-6N to	SP-CTZ-37900	MGO	1470	0	0	7	0
S04	3550-123-033-BMC-6N to	CTZ-3955C	BMC	4200	0	0	20	0
S04	3550-600-175-MGO-6N to	CRSEVEN	MGO	1260	0	0	6	0
S04	3550-600-175-MGO-6N to	CRSIX	MGO	5040	0	0	24	0
S06	3820-290-208-RMC-7S to	KOTEKA 4180B	RMC	290	0	1	1	0
S06	3835-317-040-RMC-7S to	BALI 4090P	RMC	67	1	0	0	0
S06	3835-317-040-RMC-7S to	KOTEKA 4180B	RMC	1905	2	9	5	0
S06	3865-335-165-RCW-7S to	BALI 4090P	RCW	80	0	1	0	0
S06	3865-335-165-RCW-7S to	CRUSHER-OHS3	RCW	5650	0	5	25	0
S06	3865-335-165-RCW-7S to	KDL	RCW	1500	0	3	6	0

- **Pit Graph reports:** showing VIMS events located in a mine map, including Brake temperature events and RAC events.

FPC is also used to assess the benefits of the proposed road repairs in terms of Fuel Savings and Productivity.

The impact of machine application in component life is assessed by detailed documentation of application and operation parameters to each component type (see attached file, *Component Life health – parameters and thresholds.xls*).

The attached file, *Application & Operation Process Flow.xls*, has a current “as is” process and then shows the possible “to be” processes, broken down into 100-and 200-level overviews. The 200-level process has the sub-processes:

1. **Road Selection**: objective is to select a haul road with the highest productivity and associate truck traffic for analysis.
2. **Truck Selection**: objective is to select Trucks with the representative strut pressures and assign these trucks to the selected road.
3. **Daily Data Analysis**: objective is to identify and document outstanding road defects and Operational Conditions (Road Condition Monitoring).
4. **Daily Reports**: objective is to explain Road and Operational Conditions in the last seven days and show associated benefits for the improvement action.
5. **Weekly Reports**: objective is to assess benefits of recommended repair actions.
6. **Road Repair**: objective is to eliminate identified road defects and communicate for follow up action status

7. **Operator Training**: objective is to improve operational practices according to recommended actions (weekly report).
8. **Action Follow-up**: objective is to review the weekly report and generate an action plan according to recommendations.

3.0 Implementation Steps

1. Develop a process map of "As-Is" activities associated with Road Management
2. Draft a process map of "To-Be" activities that the Dealer wants to achieve in collaboration with the Customer. This is where the RDM methodology will suggest areas to improve the process of road management.
3. Develop individual objectives for each of the sub-processes.
4. Develop the "To-Be" process map with the proposed objectives.
5. Identify which tools would be required to perform the desired process, such as:
   a. FPO: current version 1.5
   b. VIMS
   c. MineStar, Wenco or Modular: Any mine controller tool
6. Assign specific process and sub-processes to personnel in the operation (vertical axis in the "To-Be" process shown before)

Things to consider:

- Different departments involved
- On-site Equipment Maintenance
- Off-site Component Rebuilt Center
- Mine Operations
- Mine Production
- Mine Controller (Dispatcher)
- Training
- Operator Training
- Continuous Improvement: Road Defect Management assesses road condition and benefits of proposed repairs. There should be a structured and documented CI process in place to ensure that recommendations are acted upon and road defects are repaired.

4.0 Benefits

- This Best Practice focuses on the management of existing Haul Roads. A well managed haul road can deliver the following benefits to the **Customer**:
  - Better ride quality for operators and associated decrease in operator fatigue.
  - Improved productivity through better cycle times – gains of 10% are typical
  - Cost reduction through decreased tire wear and decreased fuel burn.
  - More efficient use of haul road support equipment (graders, wheel dozers etc)

- It also delivers the following benefits to the **Dealer:**
  - Reduced risk on impact in machine application with MARC rates and overall repair and maintenance costs.
  - Increased frame, suspension and power train component life due to lower stress being transferred to the truck and its components
  - Lower rolling resistance leading to improved powertrain and engine life.

- Additional benefits:
  - Improved safety through minimizing identified hazards on the road, such as excessive machine speed limits and road slippery conditions.
  - Improving Customer Satisfaction, through developing a better communication channel to discuss and analyze road defects and associated benefits if repaired
  - Profitability associated with increased productivity (Customer) with less stress to the machine (Dealer)
  - Improved perception of machine performance, through an improved machine reliability and durability.

- Lower Cost/Ton: obtained by reducing overall operating cost (fuel consumption and better use of support equipment) and improving overall productivity (reduced cycle times and better payload management)
- Operator/Technician Safety: obtained by minimizing road hazards and improving riding quality.

5.0 Resources Required

- Tools
  - Digital Still Camera
  - Inclinometer
  - Laptop (Field work) + Desktop (Office work)
  - VIMS download cable
  - FPC software
- Support equipment
  - Field utility truck (pickup truck)
  - Motor grader
  - Wheel dozer
  - Small wheel loader
  - Articulated truck (optional)
  - Water truck
- People
  - From Dealer: A dedicated person (or group of people) to analyze machine health and application. This can be achieved with the implementation of an Application and Operations crew (such as the one in Batu Hijau and Freeport), or by the Fleet Analyst.
  - From Customer: “Road Crew” or “Road Maintenance” crew to receive, execute and report back to the Dealer road repair and recommended actions.
  - On-site Equipment Maintenance
    - Application and Operations Department
    - Fleet Analyst
  - Off-site Component Rebuilt Center
    - Failure Analysis team
  - Mine Operations
    - Pit Supervisor
  - Mine Production
    - Production Supervisor
  - Mine Controller (Dispatcher)
    - Mine Controller

- Additional resources:
  - Regional Caterpillar Equipment Management Consultant for information regarding FPO.
  - A detailed Road Defect Management process is about to be released. It is currently applying for Intellectual Property rights. Once that is cleared, the detailed process map and associated software tool will be released and available to Caterpillar Dealers.
  - The “Mining Project Manager’s Toolkit” KN is also a good source of information: https://kn.cat.com/cat.cfm?id=6232
  - FPO End User Forum provides good examples of Road analysis using FPO amongst other tools: https://kn.cat.com/know.cfm?id=130516 (restricted access)

6.0 Supporting Attachments / References

- *Application & Operation Process Flow Final R1.xls*
- *Component Life health - parameters and thresholds.xls*

7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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